BME lecture week:1

I. Course Code and Title

BAE 1201: Basic Mechanical Engineering

II. Credit

Three credit hours (3 hours of theory per week)

III. Nature

Core Course for EEE

IV. Prerequisite

PHY 1101: Physics 1

V. Mission: AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB) is committed to provide quality and excellent computer-based academic programs responsive to the emerging challenges of the time. It is dedicated to nurture and produce competent world class professional imbued with strong sense of ethical values ready to face the competitive world of arts, business, science, social science and technology.

VI. Vision: AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB) envisions promoting professionals and excellent leadership catering to the technological progress and development needs of the country.

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Textbooks:

[1] R.S. Khurmi & J.K.Gupta, “A Text Book of Thermal Engineering”, two colour edition 1978(Reprint 2003), S.Chand & Company Ltd.

[2] R.S. Khurmi, “A Text Book of Hydraulics, Fluid Mechanics and Hydraulic Machines”, S.Chand & Company Ltd, Reprint 2007

References:

[1] V.P.Vasandani & D.S. Kumar, “Heat Engineering”, Metropolitan Book Co.(Pvt.) Ltd. Reprint 2001

[2] M.J. Moran & H.N. Shapiro, “Engineering Thermodynamics”, John Wiley and Sons.2000

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Course contents of first 2 weeks

Week1: *Properties of steam – internal energy, enthalpy, entropy and quality of steam, saturated and superheated steam, uses of steam table and Mollier diagram*

Week2: *Steam generating unit, boiler mountings &accessories*.

What is basic mechanical engineering?

Design, construction and production of machine and materials. Manufacturing, maintenance and control of mechanical systems. An engineering discipline dealing with physical and chemical sciences with understanding of mathematics.

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•Why we study properties of steam and steam generating units? To have knowledge on how to produce power/electricity.

•How to produce power/electricity? By using kinetic energy of high pressure and temperature steam.

•What are the processes involve in producing electricity? Converting one form of power to another form of power by means of working on the systems.

•Process in elaboration: Steam kinetic energy is attained in the boiler via fuel which run the steam turbine; kinetic energy is then converted to rotational energy in turbine. The turbine rotor-shaft is coupled with a electro-magnetic generator. Where rotational energy of the shaft converted to electro-magnetism and hence, final product electricity supplied to the grid.

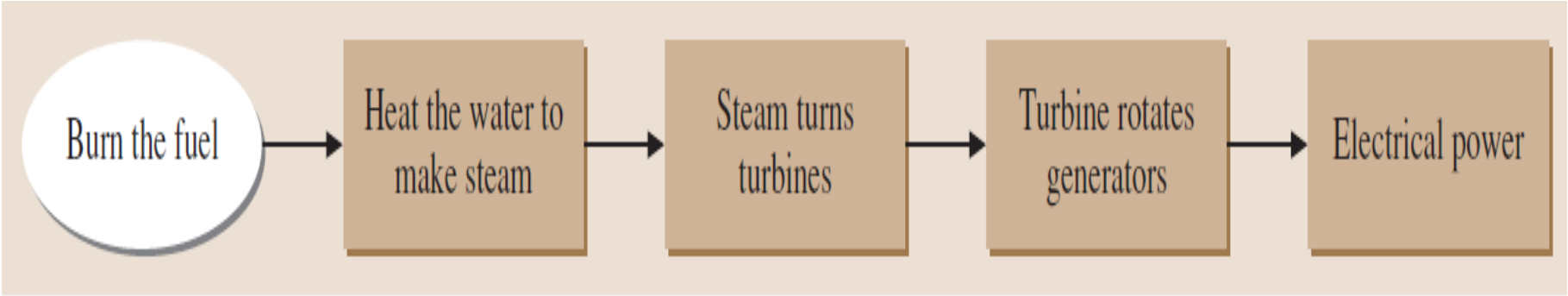


Figure: Structure of energy systems – fossil sources

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-Why steam, why not gas? We can use both.

-For what purpose we use steam/gas? As a working fluid.

-What steam means to us? Is the gas/vapor phase of water! (How about fog/cloud/vapour)?

-What are the steam properties? The thermodynamic properties. Such as: enthalpy (H), entropy (S), energy (E), work (W).

-What is enthalpy; entropy; and internal energy? Thermodynamic functions (The characteristic of a system by which it can be identified and differentiated from other systems). For an example, thermodynamic functions of steam are shown in steam table and Mollier diagram (thermodynamic tools), discussed later.

Enthalpy of a system indicates the energy which is defined as the sum of internal energy of a system and the product of the pressure and volume (flow energy or work flow) of the system. Also, the change in enthalpy is the sum of the change in the internal energy and the work done. Also, enthalpy is the sum of internal energy and external energy of a matter. When we say "internal energy", that is the translational energy, rotational energy, vibrational energy and the kinetic energy of a matter. “External energy” means the external properties; pressure and volume of a matter.

$H = U + pV$ OR, (+ absorbed, -released) [Units are S.I]

$dh = TdS + Vdp$

In an Isothermal (constant temperature) process, the change in entropy, $\Delta s = q/T$

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Entropy is defined as the non-work producing form of energy. It is also regarded as the energy that is not available for performing useful work within a certain environment. Entropy is a measure of irreversibility or disorder or randomness of a system. An ordered system has low entropy. A disordered system has high entropy. Expressed by change in enthalpy divided by temperature, since temperature is a measure of enthalpy of a system. Entropy (deltaS) is also an energy in a system but it is a chaotic energy that can't be transformed into a useful energy. In other words, it's a measurement of the degree of randomness of energy in a system.

Example 1:

Case 1: An object is kept at room temperature, i.e. 298K, and by addition of heat its temperature is increased to 398K.

Case 2: There is an object at 1000K and the temperature is increased to 1100K.

In both cases, heat addition will almost be same, and so will be the enthalpy change. But in first case there is a change of .25 and in second one, just .10. So we require another term, which is entropy.

Example 2:

-Do by yourself. Calculate the volume 1 kg of vapor would occupy under the following conditions: h= 2734 kJ, u= 2550 kJ, p= 365.64 kPa= 365.64 kN/m2, V = ?

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-What we know about the thermodynamic properties? When a sample undergoes a heat treatment, it showed change in thermo physical and chemical properties. Such as: phase change, change in thermal equilibrium, degradation process.

-What are the common terminology/units to measure the change in overall properties of a system? Pressure (P), temperature (T), volume (V), change of mass (Δm).

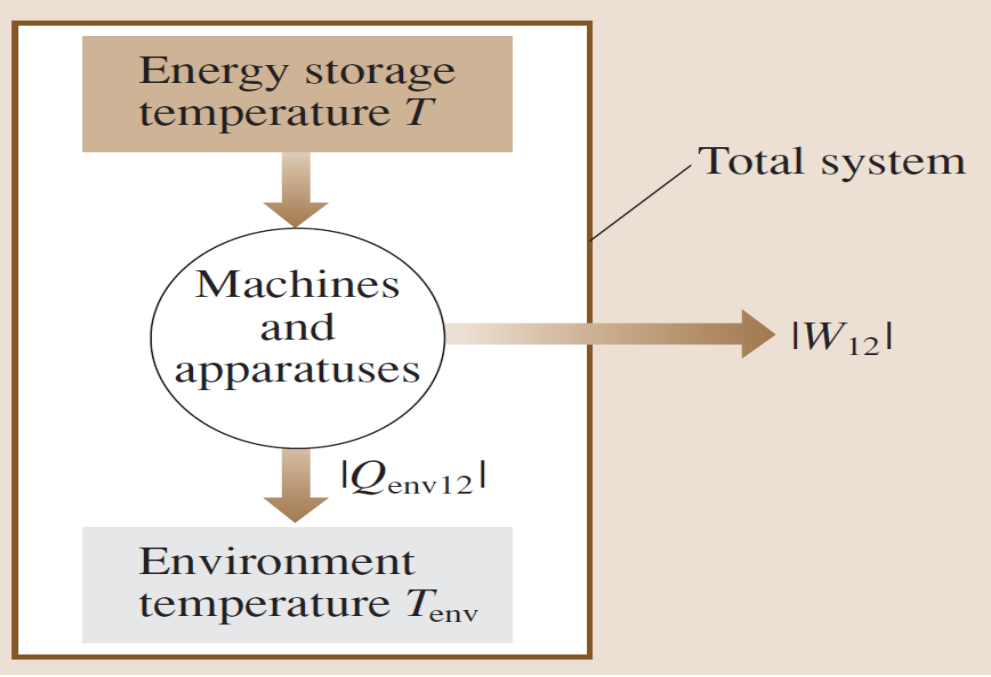


Figure: Conversion of heat into work.

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-Heat graph of steam: Steam can be visible when pure and dry (above NLM line).

-At critical point, TN= 374.15°C, PN= 221.2bar,

-Heat can affect the curve in two ways:

1. state : same but Temperature : increases.

2. state : change but Temperature : constant.

During the formation of the superheated steam, from water at freezing point, the heat is absorbed in the following three stages :

1. The heating of water up to boiling temperature or saturation temperature (t) is shown by AB in Fig. 7.2. The heat absorbed by the water is AP, known as *sensible heat* or *liquid heat* or *total heat of water*.

2. The change of state from liquid to steam is shown by BC. The heat absorbed during this stage is PQ, known as *latent heat of vaporisation*.

3. The superheating process is shown by CD. The heat absorbed during this stage is QR, known as *heat of superheat*. Line AR represents the *total heat of the superheated steam*.

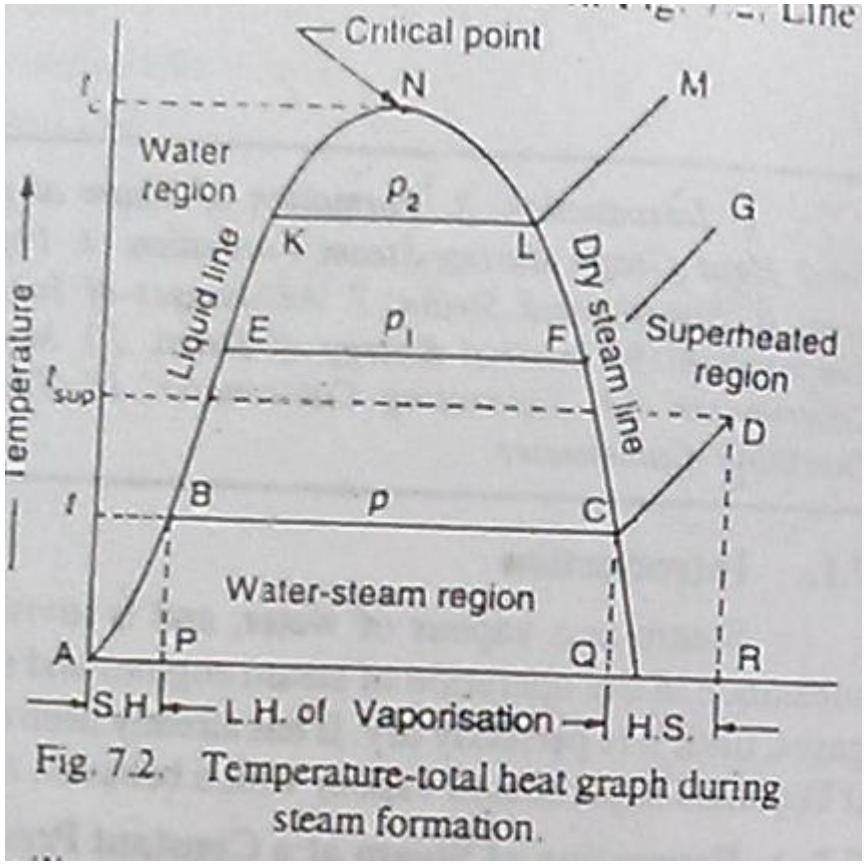


Fig. 7.2. Temperature-total heat graph during steam formation.

S.H= sensible heat, *t*=boiling or saturation temperature (AB); and heat absorbed by water/total heat/liquid heat (AP).
L.H= latent heat (state 2),
H.S= heat of superheat. Point A= initial condition of H2O at 0°C and *p* pressure.
ABCD= T and heat at specific pressure *p*.

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steam formation.

If the pressure is increased (say p_1 bar), the boiling temperature also increases. The point E represents the boiling temperature or saturation temperature at pressure p_1 and F is the point of dry saturated steam. Line FG shows the constant pressure process, in which the steam is superheated. Similarly, a family of curves may be drawn for different pressures as shown in the figure.

The line passing through the points A, B, E, K is known as *saturated liquid line* which forms boundary line between water and steam. Similarly, a line passing through dry steam points L, F, C is known as *dry saturated steam line* which forms boundary line between wet and superheated steam. Sometimes, these terms are briefly written as *liquid line* and *dry steam line*, but the word "saturated" is always understood.

It may also be noted from the figure, that when the pressure and saturation temperature increases, the latent heat of vaporisation decreases. It becomes zero at a point (N) where liquid and dry steam lines meet. This point N is known as the *critical point* and at this point, the liquid and vapour phases merge, and become identical in every respect. The temperature corresponding to critical point N is known as *critical temperature* and the pressure is known as *critical pressure*. For steam, the critical temperature is 374.15° C and critical pressure is 221.2 bar.

7.4. Important Terms for Steam

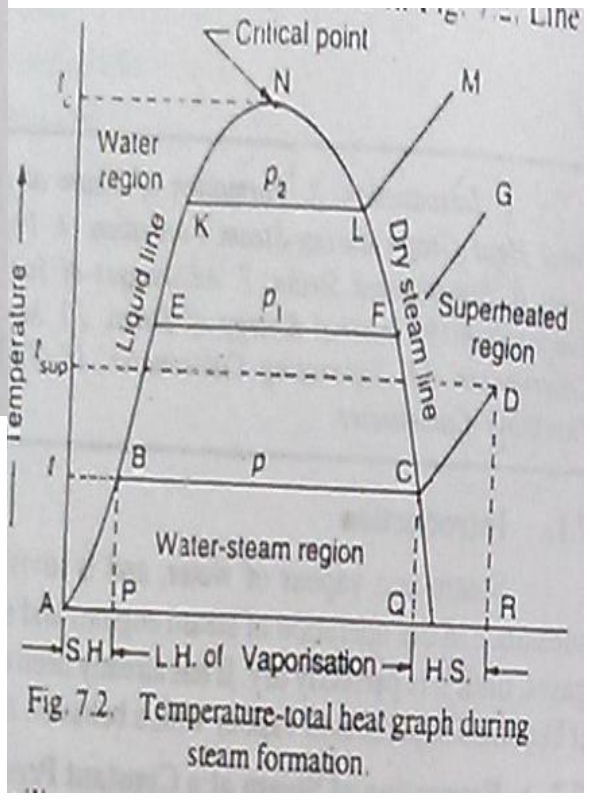


Fig. 7.2. Temperature-total heat graph during steam formation.

1. *Wet steam.* When the steam contains moisture or particles of water in suspension, it is said to be *wet steam.* It means that the evaporation of water is not complete, and the whole of the latent heat has not been absorbed.

2. *Dry saturated steam.* When the wet steam is further heated, and it does not contain any suspended particles of water, it is known as *dry saturated steam.* The dry saturated steam has absorbed its full latent heat and behaves practically, in the same way as a perfect gas.

3. *Superheated steam.* When the dry steam is further heated at a constant pressure, thus raising its temperature, it is said to be *superheated steam.* Since the pressure is constant, therefore the volume of superheated steam increases. It may be noted that the volume of one kg of superheated steam is considerably greater than the volume of one kg of dry saturated steam at the same pressure.

In actual practice, the superheated steam is produced in a separate apparatus known as superheater, so that it is out of contact with water from which it was formed.

4. *Dryness fraction or quality of wet steam.* It is the ratio of the mass of actual dry steam, to the mass of same quantity of wet steam, and is generally denoted by 'x'. Mathematically,

$$x = \frac{m_g}{m_g + m_f} = \frac{m_g}{m}$$

where m_g = Mass of actual dry steam,

m_f = Mass of water in suspension, and

m = Mass of wet steam = $m_g + m_f$

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5. *Sensible heat of water*. It is the amount of heat absorbed by 1 kg of water, when heated at a constant pressure, from the freezing point (0° C) to the temperature of formation of steam, *i.e.* saturation temperature (t). The sensible heat is also known as *liquid heat*.

The specific heat of water at constant pressure is usually taken as 4.2 kJ/kg K. Therefore heat absorbed by 1 kg of water from 0° C to t °C or sensible heat

$$= \text{Mass} \times \text{Sp. heat} \times \text{Rise in temperature}$$

$$= 1 \times 4.2\,[(t + 273) - (0 + 273)] = 4.2\,t \text{ kJ/kg}$$

Thus the sensible heat of water in kJ/kg may be obtained directly by multiplying the specific heat of water and the saturation temperature (t) in °C.

It may be noted that the sensible heat of water is taken equal to the * specific enthalpy (*i.e.* total heat) of water. It is, usually, denoted by h_f in steam tables and its value may be determined for any given pressure in kJ/kg.

6. **Latent heat of vaporisation*. It is the amount of heat absorbed to evaporate 1 kg of water, at its boiling point or saturation temperature without change of temperature. It is denoted by h_{fg} and its value depends upon the pressure. The heat of vaporisation of water or latent heat of steam is 2257 kJ/kg at atmospheric pressure.

It has been experimentally found that the value of h_{fg} decreases as the pressure increases and it is zero at critical pressure. If the steam is wet with a dryness fraction x, then the heat absorbed by it during evaporation is $x\,h_{fg}$.

7. *Enthalpy or total heat of steam*. It is amount of heat absorbed by water from freezing point to saturation temperature *plus* the heat absorbed during evaporation.

∴ Enthalpy or total heat of steam

$$= \text{Sensible heat} + \text{Latent heat}$$

It is denoted by h_g and its value for the dry saturated steam may be read directly from the steam tables. The expressions for the enthalpy of wet steam, dry steam and superheated steam are as follows :

(*i*) *Wet steam*. The enthalpy of wet steam is given by :

$$h = h_f + x\,h_{fg} \qquad \ldots (i)$$

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where x is the dryness fraction of steam.

(*ii*) *Dry steam.* We know that in case of dry steam, $x = 1$.

$$\therefore \quad h = h_g = h_f + h_{fg} \qquad \ldots (ii)$$

(*iii*) *Superheated steam.* If we further add heat to the dry steam, its temperature increases while pressure remaining constant. This increase in temperature shows the superheat stage of the steam. Thus, the total heat required for the steam to be superheated is :

h_{sup} = Total heat for dry steam + Heat for superheated steam

$$= h_f + h_{fg} + c_p (t_{sup} - t) = h_g + c_p (t_{sup} - t) \quad **$$

where c_p = Mean specific heat at constant pressure for superheated steam,

t_{sup} = Temperature of the superheated steam, and

t = Saturation temperature at the given constant pressure.

Notes : 1. The difference $(t_{sup} - t)$ is known as *degree of superheat.*

2. The value of c_p for steam lies between 1.67 kJ/kg K to 2.5 kJ/kg K.

3. *Specific volume of steam.* It is the volume occupied by the steam per unit mass at a given temperature and pressure, and is expressed in m^3/kg. It is the reciprocal of density of steam in kg/m^3. It may be noted that the value of specific volume decreases with the increase in pressure. The expressions for the volumes occupied by steam are as follows :

(*i*) *Wet steam.* Consider 1 kg of wet steam of dryness fraction x. We know that this steam will have x kg of dry steam and $(1 - x)$ kg of water. Let v_f be the volume of 1 kg of water, then

Volume of one kg of wet steam

$$= x v_g + (1 - x) v_f$$

Since v_f is very small as compared to v_g, therefore the expression $(1 - x)\, v_f$ may be neglected.

∴ Volume of 1 kg of wet steam $= x\, v_g$ m³

or specific volume of wet steam, $v = x\, v_g$ m³/kg ... (*i*)

(*ii*) *Dry steam.* We know that in case of dry steam, the mass of water in suspension is zero and dryness fraction is unity. Therefore specific volume of dry steam

$= v_g$ m³/kg ... (*ii*)

(*iii*) *Superheated steam.* We have already discussed that when the dry saturated steam is further heated under a constant pressure, there is an increase in volume with the rise in temperature. The superheated steam behaves more or less like a perfect gas. Therefore, according to Charles' law,

$$\frac{v_{sup}}{T_{sup}} = \frac{v_g}{T} \quad \text{or} \quad v_{sup} = \frac{v_g\, T_{sup}}{T} \qquad (\because\ p \text{ is constant}) \quad \ldots (iii)$$

where v_{sup} = Specific volume of superheated steam,

v_g = Specific volume of dry steam at the pressure of steam formation,

T_{sup} = Absolute temperature of superheated steam, and

T = Absolute saturation temperature at the pressure of steam formation.

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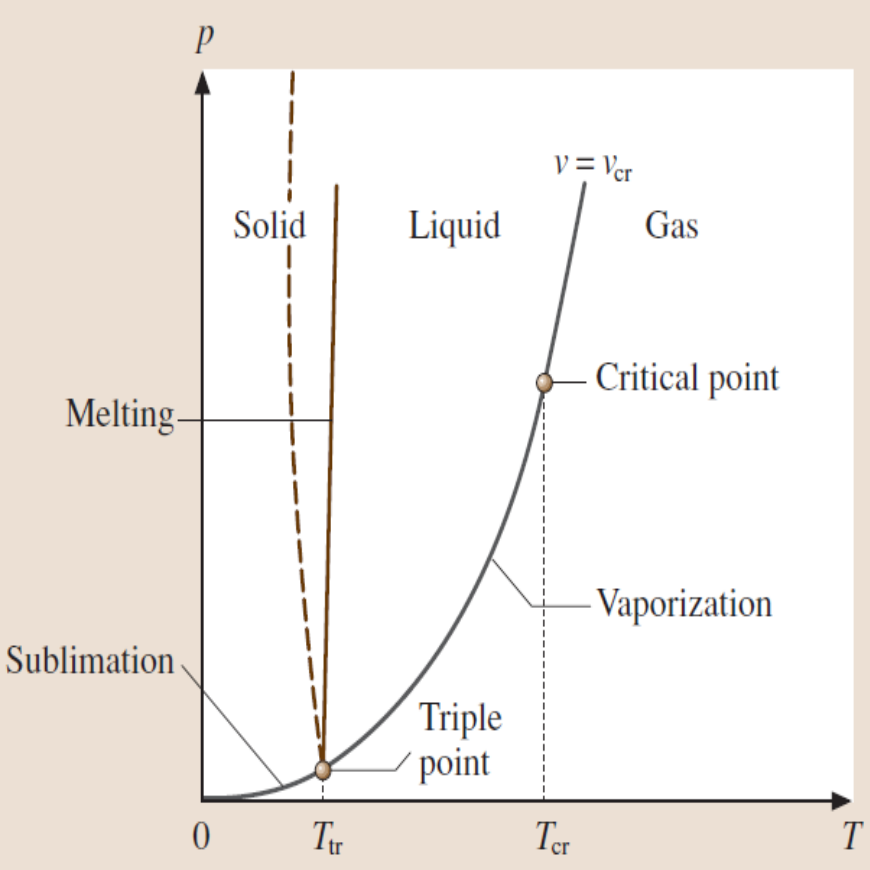


Figure: *p–T phase diagram with the three boundary lines of* the phases. (The gradient of the melting line of water is negative, *dashed line).*

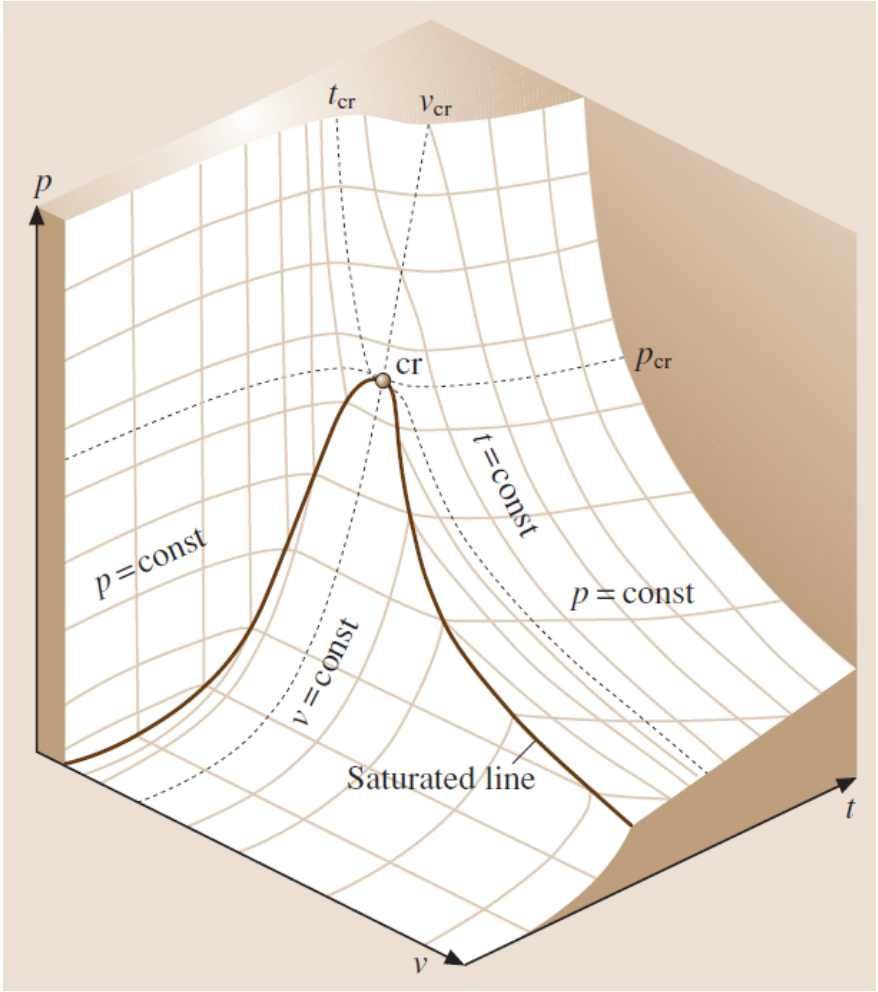


Figure: Area of states for water.

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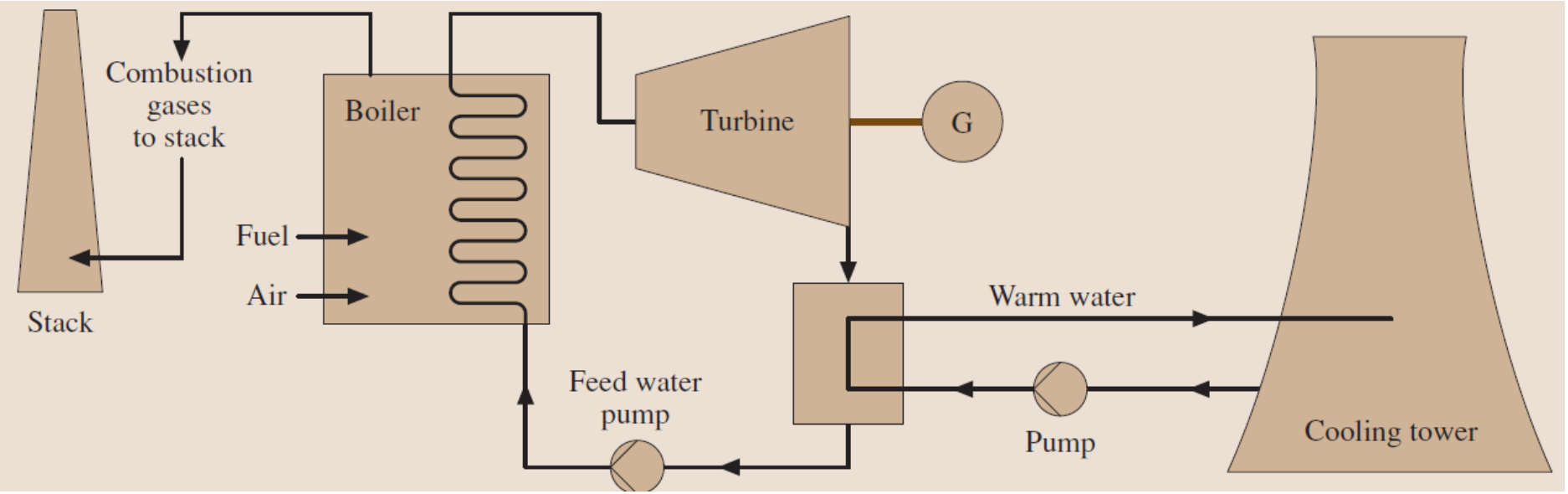


Figure: Layout of a simple Rankine cycle power plant

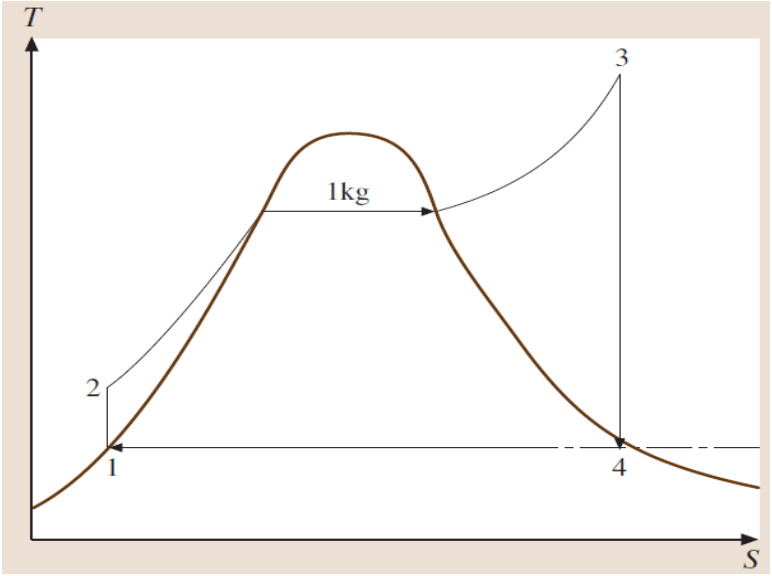


Figure: Mollier *T–S diagram of a simple Rankine cycle*

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Steam table and Mollier diagrams are almost similar thermodynamic tools for observing properties (e.g: enthalpy, entropy). There are saturated and superheated steam table by both temperature and pressure.

7.5. Steam Tables and their Uses

The properties of dry saturated steam like its temperature of formation (saturation temperature), sensible heat, latent heat of vaporisation, enthalpy or total heat, specific volume, entropy etc., vary with pressure, and can be found by experiments only. These properties have been carefully determined, and made available in a tabular form known as *steam tables*. It may be noted, that there is a slight difference in figures, quoted by various authors, and even in their various editions of a book. But in this book we shall use the standard properties which are internationally recognised.

There are two important steam tables, one in terms of absolute pressure and other in terms of temperature. An extract of the properties of dry saturated steam is shown in the following two tables :

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TABLE A-3 Properties of Saturated Water (Liquid–Vapor): Pressure Table

Press. bar	Temp. °C	Specific Volume m^3/kg Sat. Liquid $v_f \times 10^3$	Specific Volume m^3/kg Sat. Vapor v_g	Internal Energy kJ/kg Sat. Liquid u_f	Internal Energy kJ/kg Sat. Vapor u_g	Enthalpy kJ/kg Sat. Liquid h_f	Enthalpy kJ/kg Evap. h_{fg}	Enthalpy kJ/kg Sat. Vapor h_g	Entropy kJ/kg · K Sat. Liquid s_f	Entropy kJ/kg · K Sat. Vapor s_g	Press. bar
0.04	28.96	1.0040	34.800	121.45	2415.2	121.46	2432.9	2554.4	0.4226	8.4746	0.04
0.06	36.16	1.0064	23.739	151.53	2425.0	151.53	2415.9	2567.4	0.5210	8.3304	0.06
0.08	41.51	1.0084	18.103	173.87	2432.2	173.88	2403.1	2577.0	0.5926	8.2287	0.08
0.10	45.81	1.0102	14.674	191.82	2437.9	191.83	2392.8	2584.7	0.6493	8.1502	0.10
0.20	60.06	1.0172	7.649	251.38	2456.7	251.40	2358.3	2609.7	0.8320	7.9085	0.20
0.30	69.10	1.0223	5.229	289.20	2468.4	289.23	2336.1	2625.3	0.9439	7.7686	0.30
0.40	75.87	1.0265	3.993	317.53	2477.0	317.58	2319.2	2636.8	1.0259	7.6700	0.40
0.50	81.33	1.0300	3.240	340.44	2483.9	340.49	2305.4	2645.9	1.0910	7.5939	0.50
0.60	85.94	1.0331	2.732	359.79	2489.6	359.86	2293.6	2653.5	1.1453	7.5320	0.60
0.70	89.95	1.0360	2.365	376.63	2494.5	376.70	2283.3	2660.0	1.1919	7.4797	0.70
0.80	93.50	1.0380	2.087	391.58	2498.8	391.66	2274.1	2665.8	1.2329	7.4346	0.80
0.90	96.71	1.0410	1.869	405.06	2502.6	405.15	2265.7	2670.9	1.2695	7.3949	0.90
1.00	99.63	1.0432	1.694	417.36	2506.1	417.46	2258.0	2675.5	1.3026	7.3594	1.00
1.50	111.4	1.0528	1.159	466.94	2519.7	467.11	2226.5	2693.6	1.4336	7.2233	1.50
2.00	120.2	1.0605	0.8857	504.49	2529.5	504.70	2201.9	2706.7	1.5301	7.1271	2.00
2.50	127.4	1.0672	0.7187	535.10	2537.2	535.37	2181.5	2716.9	1.6072	7.0527	2.50
3.00	133.6	1.0732	0.6058	561.15	2543.6	561.47	2163.8	2725.3	1.6718	6.9919	3.00
3.50	138.9	1.0786	0.5243	583.95	2546.9	584.33	2148.1	2732.4	1.7275	6.9405	3.50
4.00	143.6	1.0836	0.4625	604.31	2553.6	604.74	2133.8	2738.6	1.7766	6.8959	4.00
4.50	147.9	1.0882	0.4140	622.25	2557.6	623.25	2120.7	2743.9	1.8207	6.8565	4.50
5.00	151.9	1.0926	0.3749	639.68	2561.2	640.23	2108.5	2748.7	1.8607	6.8212	5.00
6.00	158.9	1.1006	0.3157	669.90	2567.4	670.56	2086.3	2756.8	1.9312	6.7600	6.00
7.00	165.0	1.1080	0.2729	696.44	2572.5	697.22	2066.3	2763.5	1.9922	6.7080	7.00
8.00	170.4	1.1148	0.2404	720.22	2576.8	721.11	2048.0	2769.1	2.0462	6.6628	8.00
9.00	175.4	1.1212	0.2150	741.83	2580.5	742.83	2031.1	2773.9	2.0946	6.6226	9.00
10.0	179.9	1.1273	0.1944	761.68	2583.6	762.81	2015.3	2778.1	2.1387	6.5863	10.0
15.0	198.3	1.1539	0.1318	843.16	2594.5	844.84	1947.3	2792.2	2.3150	6.4448	15.0
20.0	212.4	1.1767	0.09963	906.44	2600.3	908.79	1890.7	2799.5	2.4474	6.3409	20.0
25.0	224.0	1.1973	0.07998	959.11	2603.1	962.11	1841.0	2803.1	2.5547	6.2575	25.0
30.0	233.9	1.2165	0.06668	1004.8	2604.1	1008.4	1795.7	2804.2	2.6457	6.1869	30.0
35.0	242.6	1.2347	0.05707	1045.4	2603.7	1049.8	1753.7	2803.4	2.7253	6.1253	35.0
40.0	250.4	1.2522	0.04978	1082.3	2602.3	1087.3	1714.1	2801.4	2.7964	6.0701	40.0
45.0	257.5	1.2692	0.04406	1116.2	2600.1	1121.9	1676.4	2798.3	2.8610	6.0199	45.0
50.0	264.0	1.2859	0.03944	1147.8	2597.1	1154.2	1640.1	2794.3	2.9202	5.9734	50.0
60.0	275.6	1.3187	0.03244	1205.4	2589.7	1213.4	1571.0	2784.3	3.0267	5.8892	60.0
70.0	285.9	1.3513	0.02737	1257.6	2580.5	1267.0	1505.1	2772.1	3.1211	5.8133	70.0
80.0	295.1	1.3842	0.02352	1305.6	2569.8	1316.6	1441.3	2758.0	3.2068	5.7432	80.0
90.0	303.4	1.4178	0.02048	1350.5	2557.8	1363.3	1378.9	2742.1	3.2858	5.6772	90.0
100.	311.1	1.4524	0.01803	1393.0	2544.4	1407.6	1317.1	2724.7	3.3596	5.6141	100.
110.	318.2	1.4886	0.01599	1433.7	2529.8	1450.1	1255.5	2705.6	3.4295	5.5527	110.

Example 7.1. *Calculate the enthalpy of 1 kg of steam at a pressure of 8 bar and dryness fraction of 0.8. How much heat would be required to raise 2 kg of this steam from water at 20° C?*

Solution. Given : $p = 8$ bar ; $x = 0.8$

Enthalpy of 1 kg of steam

From steam tables, corresponding to a pressure of 8 bar, we find that

$$h_f = 720.9 \text{ kJ/kg} \quad \text{and} \quad h_{fg} = 2046.5 \text{ kJ/kg}$$

We know that enthalpy of 1 kg of wet steam,

$$h = h_f + x\, h_{fg} = 720.9 + 0.8 \times 2046.5 = 2358.1 \text{ kJ} \quad \textbf{Ans.}$$

Heat required to raise 2 kg of this steam from water at 20° C

We have calculated above the enthalpy or total heat required to raise 1 kg of steam from water at 0° C. Since the water, in this case, is already at 20° C, therefore

Heat already in water $= 4.2 \times 20 = 84$ kJ

∴ Heat required per kg of steam

$$= 2358.1 - 84 = 2274.1 \text{ kJ}$$

and heat required for 2 kg of steam

$$= 2 \times 2274.1 = 4548.2 \text{ kJ} \quad \textbf{Ans.}$$

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Example 7.2. *Determine the quantity of heat required to produce 1 kg of steam at a pressu of 6 bar at a temperature of 25° C, under the following conditions :*

1. when the steam is wet having a dryness fraction 0.9 ,

2. when the steam is dry saturated , and

3. when it is superheated at a constant pressure at 250° C assuming the mean specific heat superheated steam to be 2.3 kJ/kg K.

Solution. Given : $p = 6$ bar ; $t_w = 25°$ C , $x = 0.9$; $t_{sup} = 250°$ C ; $c_p = 2.3$ kJ/kg K

From steam tables, corresponding to a pressure of 6 bar, we find that

$$h_f = 670.4 \text{ kJ/kg} ; \quad h_{fg} = 2085 \text{ kJ/kg} \text{ and } t = 158.8° \text{ C}$$

1. *When the steam is wet*

We know that enthapy or total heat of 1 kg of wet steam,

$$h = h_f + x\, h_{fg} = 670.4 + 0.9 \times 2085 = 2546.9 \text{ kJ}$$

Since the water is at a temperature of 25° C, therefore

Heat already in water $= 4.2 \times 25 = 105$ kJ

∴ Heat actually required $= 2546.9 - 105 = 2441.9$ kJ **Ans.**

2. *When the steam is dry saturated*

We know that enthalpy or total heat of 1 kg of dry saturated steam,

$$h_g = h_f + h_{fg} = 670.4 + 2085 = 2755.4 \text{ kJ}$$

∴ Heat actually required $= 2755.4 - 105 = 2650.4$ kJ **Ans.**

3. *When the steam is superheated*

We know that enthalpy or total heat of 1 kg of superheated steam,

$$h_{sup} = h_g + c_p (t_{sup} - t) = 2755.4 + 2.3\,(250 - 158.8) = 2965.16 \text{ kJ}$$

∴ Heat actually required $= 2965.16 - 105 = 2860.16$ kJ **Ans.**

Example 7.3. *Determine the condition of steam in the following cases*

1. At a pressure of 10 bar and temperature 200° C.

2. At a pressure of 10 bar and volume 0.175 m^3/kg.

Solution. Given $p = 10$ bar, $t = 200°$ C, $v = 0.175$ m^3/kg

1. *Condition of steam at temperature of 200° C*

From steam tables, corresponding to a pressure of 10 bar, we find that

$$v_g = 0.194 \text{ m}^3/\text{kg}; \; h_f = 762.6 \text{ kJ/kg}; \text{ and } t = 179.9° \text{C}$$

Since the saturation temperature at 10 bar is (179.9° C) or lower than the given temperature of the steam (200° C), therefore the given steam is superheated. The degree of superheat

$$= 200 - 179.9 = 20.1° \text{C} \quad \textbf{Ans.}$$

2. *Condition of steam at a volume of 0.175 m^3/kg*

Since the volume of given steam (0.175 m^3/kg) is less than the specific volume of the dry saturated steam (0.194 m^3/kg), therefore the given steam is wet. The dryness fraction of steam,

$$x = \frac{0.175}{0.194} = 0.902 \quad \textbf{Ans.}$$

Example 7.4. *Steam enters an engine at a pressure of 12 bar with a 67° C of superheat. It is exhausted at a pressure of 0.15 bar and 0.95 dry. Find the drop in enthalpy of the steam.*

Solution. Given : $p_1 = 12$ bar ; $t_{sup} - t = 67°$ C ; $p_2 = 0.15$ bar ; $x = 0.95$

From steam tables, corresponding to a pressure of 12 bar, we find that

$$h_f = 798.4 \text{ kJ/kg}, \quad h_{fg} = 1984.3 \text{ kJ/kg}$$

We know that enthalpy or total heat of 1 kg of superheated steam,

$$h_{sup} = h_f + h_{fg} + c_p (t_{sup} - t) = 798.4 + 1984.3 + 2 \times 67$$

$$= 2916.7 \text{ kJ/kg} \quad \textbf{Ans.} \qquad \text{(Taking } c_p = 2 \text{ kJ/kg K)}$$

Similarly, from steam tables, corresponding to a pressure of 0.15 bar, we find that

$$h_f = 226 \text{ kJ/kg}, \quad h_{fg} = 2373.2 \text{ kJ/kg}$$

We know that enthalpy or total heat of 1 kg of wet steam,

$$h = h_f + x h_{fg} = 226 + 0.95 \times 1984.3 = 2111 \text{ kJ/kg}$$

∴ Drop in enthalpy of the steam

$$= h_{sup} - h = 2916.7 - 2111 = 805.7 \text{ kJ/kg} \quad \textbf{Ans.}$$

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Example 7.5. *A steam engine obtains steam from a boiler at a pressure of 15 bar and 0.98 dry. It was observed that the steam loses 21 kJ of heat per kg as it flows through the pipe line, pressure remaining constant. Calculate dryness fraction of the steam, at the engine end of the pipeline.*

Solution. Given : $p = 15$ bar , $x = 0.98$; Heat loss = 21 kJ/kg

From steam tables, corresponding to a pressure of 15 bar, we find that

$$h_f = 844.6 \text{ kJ/kg} , \quad h_{fg} = 1945.3 \text{ kJ/kg}$$

We know that enthalpy of wet steam at the boiler end,

$$h_1 = h_f + x h_{fg} = 844.6 + 0.98 \times 1945.3 = 2751 \text{ kJ/kg}$$

Since the steam loses 21 kJ/kg of steam, therefore enthalpy of wet steam at the engine end,

$$h_2 = 2751 - 21 = 2730 \text{ kJ}$$

Let x_2 = Dryness fraction of steam at the engine end.

Since the pressure remains constant, therefore h_f and h_{fg} is same. We know that

$$h_2 = h_f + x_2 h_{fg}$$

$$2730 = 844.6 + x_2 \times 1945.3 \text{ or } x_2 = 0.97 \text{ Ans.}$$

Example 7.6. *Determine the volume of 1 kg of superheated steam at a pressure of 20 bar and a temperature of 300° C.*

Solution. Given : $p = 20$ bar ; $T_{sup} = 300°\text{ C} = 300 + 273 = 573$ K

Let v_{sup} = Volume of 1 kg of superheated steam.

From steam tables, corresponding to a pressure of 20 bar, we find that

$$t = 212.4°\text{ C} \text{ and } v_g = 0.1 \text{ m}^3/\text{kg}$$

$$T = 212.4 + 273 = 485.4 \text{ K}$$

We know that the superheated steam obeys Charles's law.

$$\therefore \quad \frac{v_g}{T} = \frac{v_{sup}}{T_{sup}} \qquad \text{(} p \text{ is constant)}$$

or

$$v_{sup} = \frac{v_g}{T} \times T_{sup} = \frac{0.1}{485.4} \times 573 = 0.118 \text{ m}^3 \textbf{ Ans.}$$

Example 7.7. *A boiler is supplied with feed water at a temperature of 45° C. The water is converted into steam at a pressure of 5.5 bar and a temperature of 188° C. Determine the quantity of heat supplied per kg of steam. Assume suitable data.*

Example 7.7. *A boiler is supplied with feed water at a temperature of 45° C. The water is converted into steam at a pressure of 5.5 bar and a temperature of 188° C. Determine the quantity of heat supplied per kg of steam. Assume suitable data.*

Solution. Given : $t_w = 45°\,C$, $p = 5.5$ bar ; $t_{sup} = 188°\,C$

From steam tables, corresponding to a pressure of 5.5 bar, we find that

$$h_f = 655.8 \text{ kJ/kg} ; \; h_{fg} = 2095.9 \text{ kJ/kg} ; \; t = 155.5°\,C$$

Since the temperature of the given steam (188° C) is higher than the saturation temperature (155.5° C), therefore the steam produced is superheated. Now let us assume the value of c_p for superheated steam as 2.1 kJ/kg K. We know that enthalpy or total heat of superheated steam,

$$h_{sup} = h_f + h_{fg} + c_p (t_{sup} - t)$$

$$= 655.8 + 2095.9 + 2.1 (188 - 155.5) = 2820 \text{ kJ/kg}$$

Since the water is supplied at 45° C, therefore heat already present in the feed water

$$= 4.2 \times 45 = 189 \text{ kJ}$$

∴ **Heat actually required** = 2820 – 189 = 2631 kJ/kg **Ans.**

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Mollier diagram is named after Richard Mollier (1863-1935), a German professor who pioneered experimental research on thermodynamics associated with water, steam and water-vapor mixture. Mollier diagram is a graphical representation of a functional relationship between enthalpy, entropy, temperature, pressure and quality of steam. Mollier is often referred to as Enthalpy – Entropy Diagram or Enthalpy – Entropy Chart. They used commonly in the design and analysis associated with power plants, steam turbines, compressors, and refrigeration systems.

The constant pressure and constant temperature lines in the Mollier diagram are referred to as isobars and isotherms, respectively. In addition, the graph includes lines representing constant steam quality, "x," in the bottom half of the diagram. The bold line, spanning from left to right, in the lower half of Mollier diagram is the saturation line. The saturation line, labelled as x = 1, represents the set of points on Mollier diagram where the steam is 100% vapor. All points above the saturation line are in the superheated steam territory. All points below the saturation line represent a mixture of liquid and vapor phases.

Example 3: Find out the enthalpy (H) at 220°C (T) from steam table by interpolation.

The values of T and H at 0.1 MPa, let's assume we have following values from steam table

At constant pressure (MPa)	Temperature (°C)	Enthalpy (kJ/kg)
0.01	200	2879.6
0.01	220	H=?
0.01	250	2977.4

Now using interpolation method we can derived the value at 220C from enthalpies given at 200°C and 250°C.

$$\frac{H-H_{low}}{H_{high}-H_{low}} = \frac{T_H-T_{low}}{T_{high}-T_{low}}$$

$$H = \left(\frac{T_H-T_{low}}{T_{high}-T_{low}}\right)(H_{high} - H_{low}) + (H_{low})$$

$$H = \frac{220-200}{250-200}(2977.4 - 2879.6) + 2879.6$$

= 0.04(97.8)+2879.6=2918.7kj/kg ; this is enthalpy/H at 220°C (ANS).

is 310 – 40 = 270° C.

8.9. Enthalpy-Entropy (*h-s*) Diagram for Water and Steam or Mollier Chart

It is a graphical representation of the steam tables, in which the enthalpy (*h*) is plotted along the ordinate and the entropy (*s*) along abscissa. First of all, enthalpy and entropy of water and dry saturated steam, for any particular pressure, are obtained from the steam tables. These values of enthalpies and entropies are plotted and then liquid line and dry saturated line is obtained. Both these lines meet at C, *i.e.* the critical point as shown in Fig. 8.5. The critical point corresponds to the enthalpy of liquid and dry saturated steam at 221.2 bar.

The enthalpy-entropy chart, like temperature-entropy chart, is also very useful in solving the problems on isentropic expansion and compression of steam. In actual diagram, abscissa of the

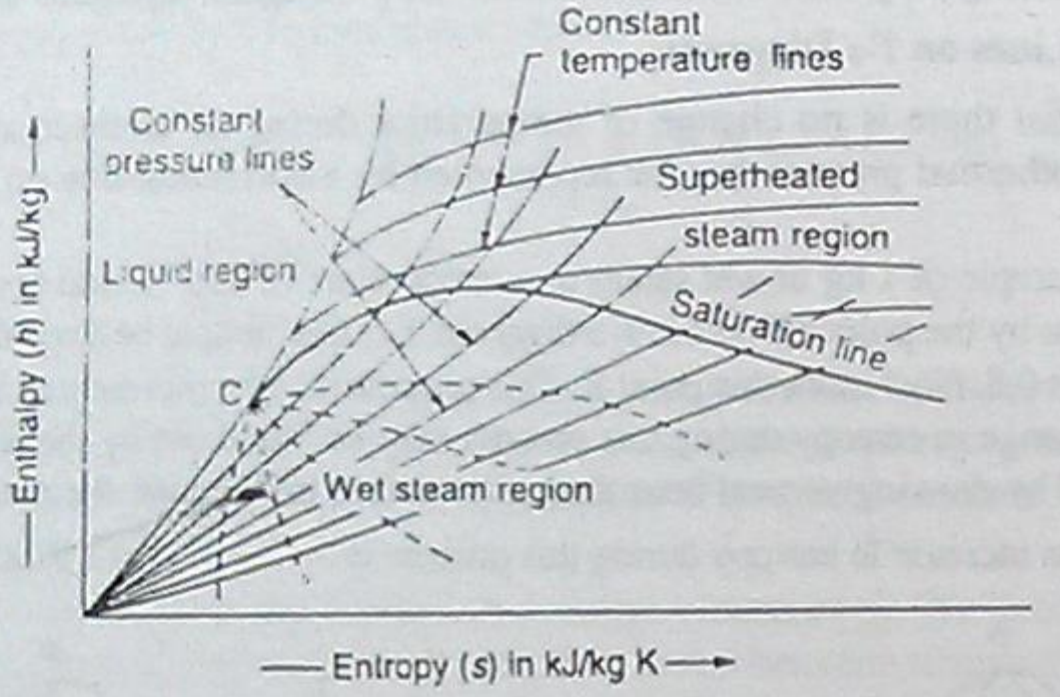


Fig. 8.5. *h-s* diagram for water and steam.

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diagram represents the entropy of 1 kg of water and steam (*i.e.* specific entropy) above the freezing point of water. The vertical ordinate shows the values of specific enthalpy (*i.e.* total heat) as shown in Fig. 8.6.

The diagram is divided into two portions, by a line termed as *saturation line*. In the lower (*i.e.* wet) region, the temperature of steam remains constant at a given pressure. In the upper (*i.e.* super heat) region the temperature of steam increases at the given pressure.

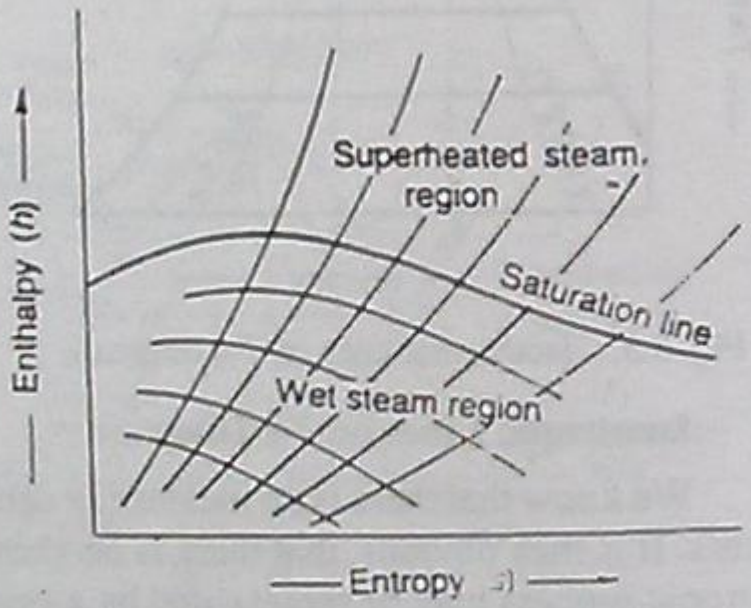


Fig. 8.6. *h-s* diagram of Mollier chart

The Mollier diagram has the following lines :

1. Dryness fraction lines,
2. Constant volume lines,
3. Constant pressure lines,
4. Isothermal lines,
5. Isentropic lines, and
6. Throttling lines. = Isenthalpic lines

These lines are discussed, in detail, in the following pages.

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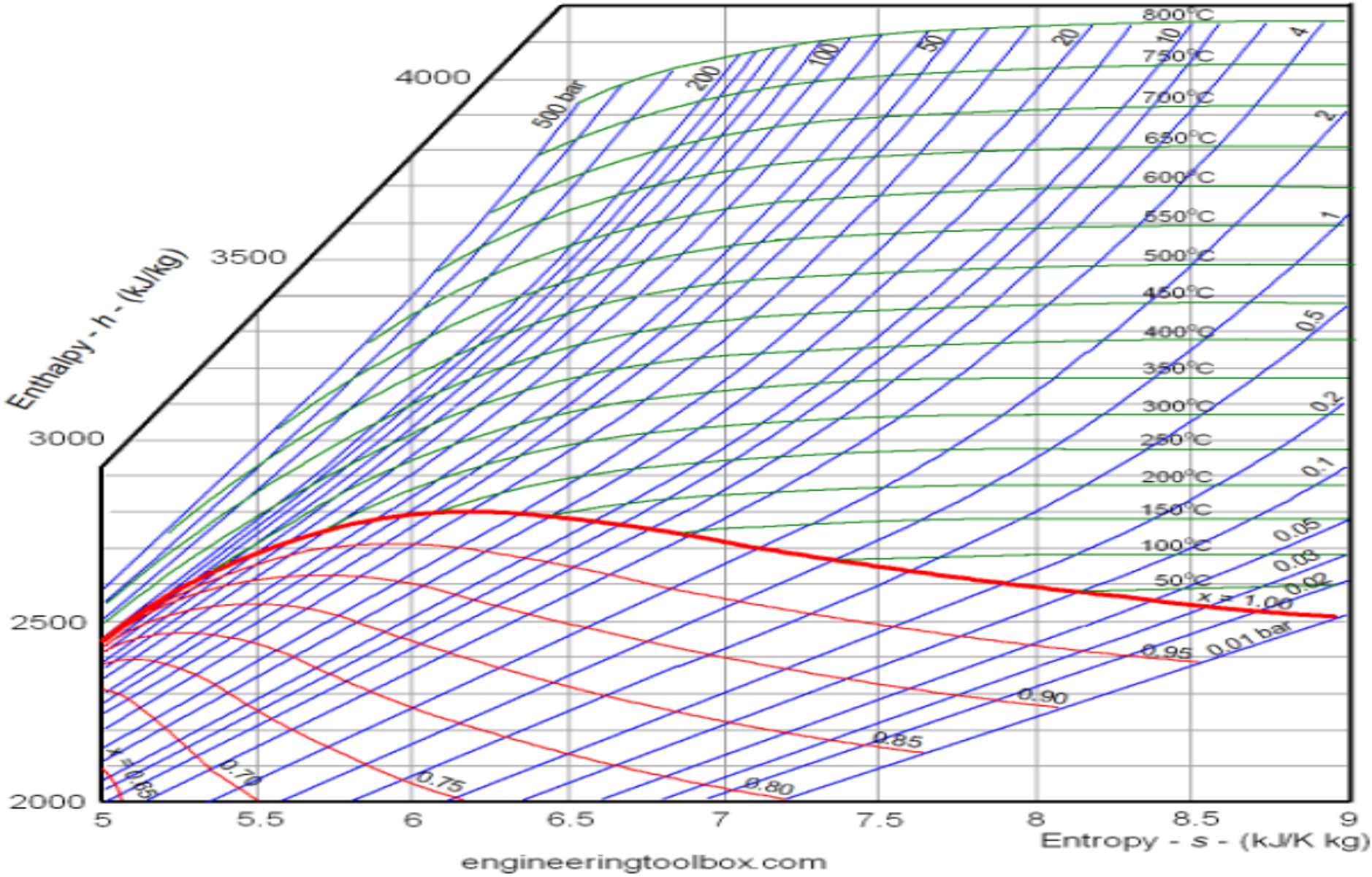


Fig: Mollier Diagram for Water-Steam

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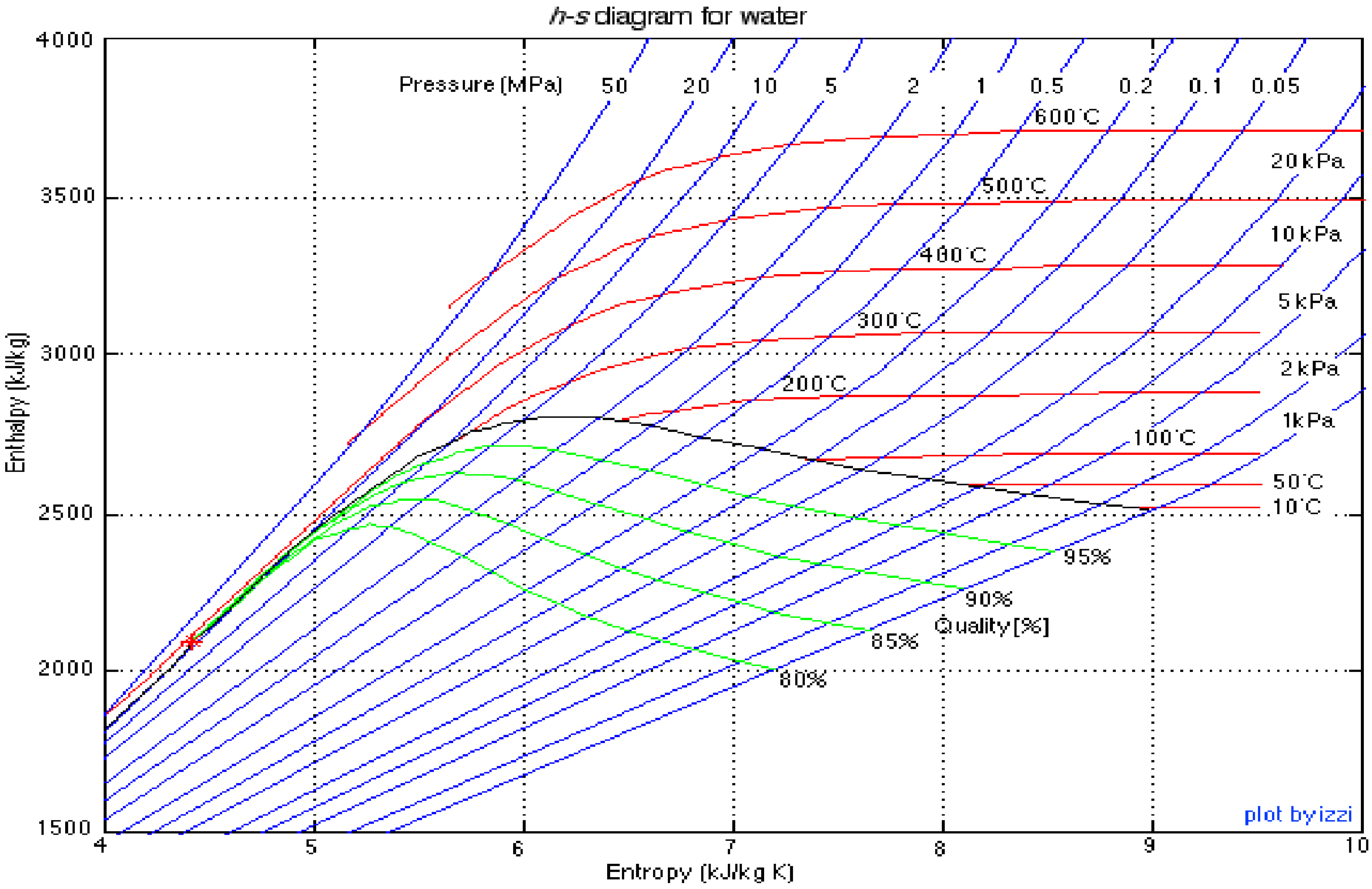


Figure: Mollier *H–S diagram*

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Example 8.4. *Using Mollier chart, find enthalpy drop and final condition of steam when it is expanded isentropically from an initial pressure of 30 bar and 350° C to a pressure of 1 bar.*

Solution. Given : $p_1 = 30$ bar ; $t_1 = 350°$ C ; $p_2 = 1$ bar

1. First of all, on the Mollier chart mark a point *A*, where the pressure line through p_1 (*i.e.* 30 bar) and temperature line through t_1 (*i.e.* 350° C) meets as shown in Fig. 8.13.

2. Since the steam is expanded isentropically (*i.e.* at a constant entropy), therefore draw a vertical line through *A* to meet the pressure line p_2 (*i.e.* 1 bar) at point *B* as shown in Fig. 8.13.

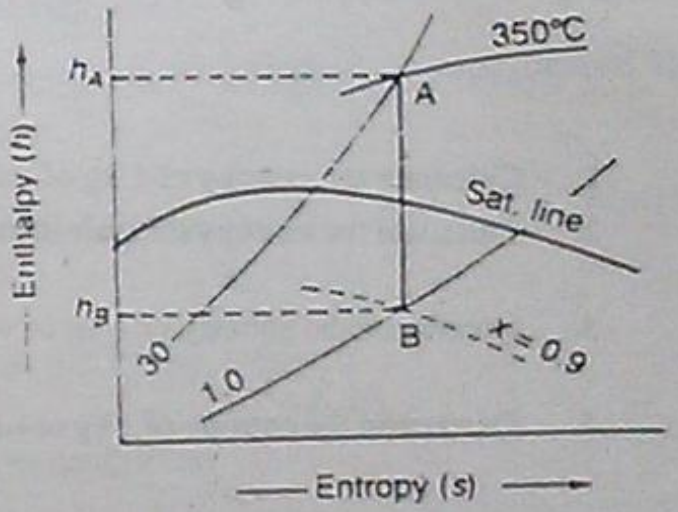


Fig. 8.13

Enthalpy drop during the process

From the Mollier chart, we find that enthalpy at *A*,

$$h_A = 3120 \text{ kJ/kg}$$

and enthalpy at *B*, $h_B = 2450$ kJ/kg

∴ Enthalpy drop during the process,

$$h = h_A - h_B = 3120 - 2450 = 670 \text{ kJ/kg}$$ **Ans.**

Final condition of the steam

From the Mollier chart, we also find that final condition (*i.e.* dryness fraction) of the steam at *B* is 0.9. **Ans.**

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is 0.9. **Ans.**

Example 8.5. *A sample of steam at 200° C and a pressure of 7 bar is first expanded at a constant enthalpy to 3 bar, and then at constant entropy to 0.5 bar. With the help of Mollier diagram, determine the change of entropy and enthalpy during the processes.*

Solution. Given $t_1 = 200°$ C ; $p_1 = 7$ bar ; $p_2 = 3$ bar ; $p_3 = 0.5$ bar

1. First of all, on the Mollier diagram mark point A where the pressure line through p_1 (*i.e.* 7 bar) and temperature line through t_1 (*i.e.* 200° C) meets as shown in Fig. 8.14.

2. Since the steam is first expanded at constant enthalpy, therefore draw a horizontal line through A to meet the pressure line through p_2 (*i.e.* 3 bar) at point *B* as shown in Fig. 8.14.

4, 3, 0.5 bar and 200C

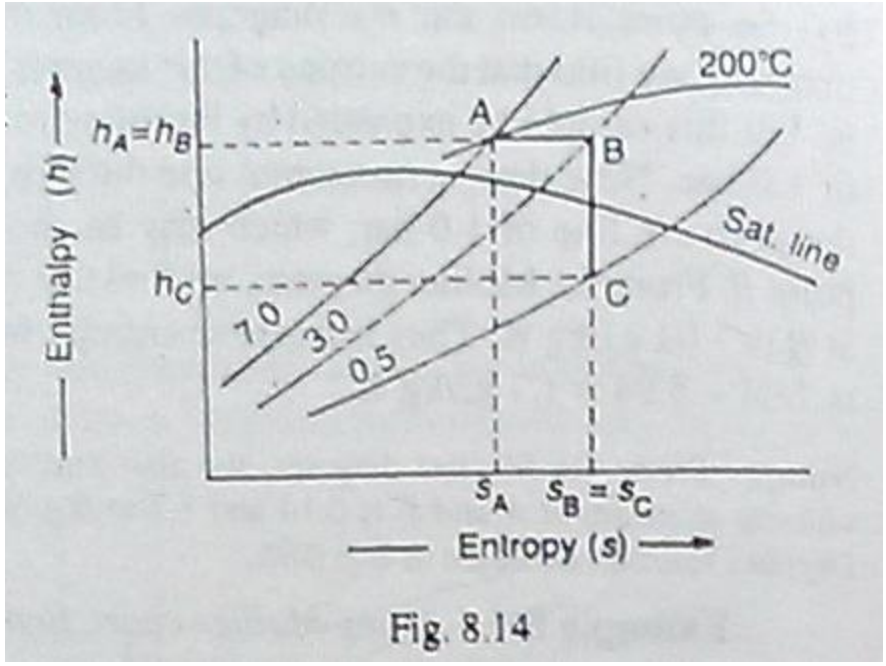


Fig. 8.14

3. Now as the steam is expanded at constant entropy, therefore draw a vertical line through B to meet the pressure line p_1 (*i.e.* 0.5 bar) at point C as shown in Fig. 8.14.

From the Mollier diagram, we find that

$h_A = h_B = 2850$ kJ/kg ; $s_A = 6.84$ kJ/kg K

$h_C = 2535$ kJ/kg ; $s_B = s_C = 7.24$ kJ/kg K

Change of entropy and enthalpy during constant enthalpy process

We know that change of entropy

$= s_B - s_A = 7.24 - 6.84$

$= 0.4$ kJ/kg K **Ans.**

and change of enthalpy = 0 **Ans.**

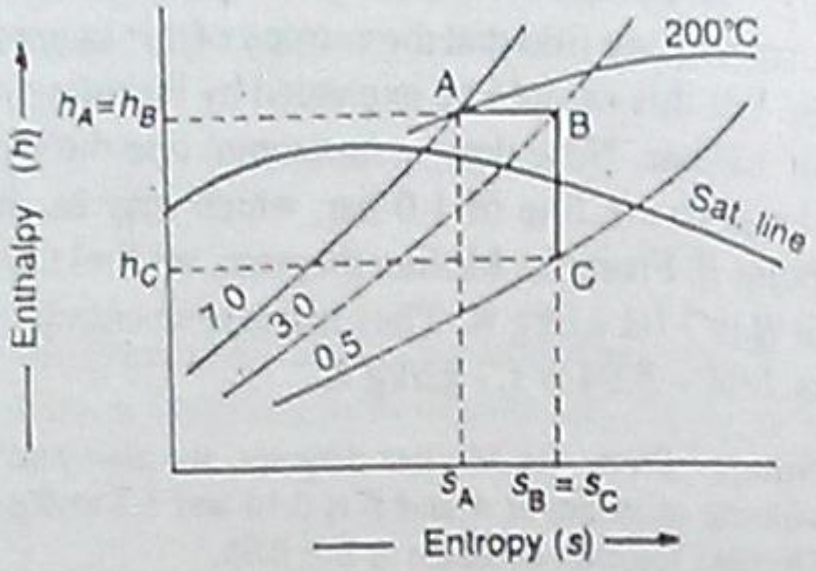


Fig. 8.14

Change of entropy and enthalpy during constant entropy process

We know that change of entropy

= 0 **Ans.**

and change of enthalpy = $h_B - h_C = 2850 - 2535 = 315$ kJ/kg **Ans.**

Thank you for your attention